

Wi-Fi Training Program

Module 3 – Assignment (Questions & Answers)

1. What are the different 802.11 PHY layer standards? Compare their characteristics.

Answer:

The IEEE 802.11 family defines various PHY (Physical Layer) standards, each improving upon the previous in terms of frequency, speed, and technology used.

Standard	Wi-Fi Gen	Frequency	Max Data Rate	Modulation	Channel Width	Key Feature
802.11 (legacy)	-	2.4 GHz	2 Mbps	DSSS / FHSS	22 MHz	First standard (1997)
802.11b	Wi-Fi 1	2.4 GHz	11 Mbps	DSSS (CCK)	22 MHz	First widely adopted
802.11a	Wi-Fi 2	5 GHz	54 Mbps	OFDM	20 MHz	Less interference at 5 GHz
802.11g	Wi-Fi 3	2.4 GHz	54 Mbps	OFDM	20 MHz	Backward compatible with 802.11b
802.11n	Wi-Fi 4	2.4 / 5 GHz	600 Mbps	OFDM + MIMO	20/40 MHz	Introduced MIMO
802.11ac	Wi-Fi 5	5 GHz	3.5 Gbps	OFDM + MU-MIMO	20–160 MHz	MU-MIMO, 256-QAM
802.11ax	Wi-Fi 6/6E	2.4/5/6 GHz	9.6 Gbps	OFDMA + MU-MIMO	20–160 MHz	OFDMA, BSS Coloring, TWT
802.11be	Wi-Fi 7	2.4/5/6 GHz	46 Gbps	OFDMA + MU-MIMO	Up to 320 MHz	Multi-Link Operation (MLO)

Key Takeaway: Each generation improves frequency efficiency, data rate, and spatial multiplexing. The move from DSSS to OFDM and then OFDMA has been the most significant technological leap.

2. What are DSSS and FHSS? How do they work?

Answer:

DSSS – Direct Sequence Spread Spectrum:

DSSS spreads the data signal across a wider frequency band by multiplying it with a high-rate chipping code (called a Barker code or similar pseudo-random sequence). Each data bit is represented by multiple chips (e.g., 11 chips per bit in 802.11b).

- The receiver uses the same chipping code to de-spread the signal and recover the original data.
- Spreading the signal makes it more resistant to narrowband interference — if interference affects part of the spectrum, the rest still carries the data.
- Used in: 802.11 (legacy) and 802.11b.

- Advantage: More robust to interference and multipath fading.
- Disadvantage: Less spectrally efficient than OFDM.

FHSS – Frequency Hopping Spread Spectrum:

FHSS transmits data by rapidly hopping the carrier frequency across multiple channels in a pseudo-random sequence known to both transmitter and receiver.

- The transmitter and receiver synchronize their hopping patterns.
- If a particular frequency is jammed or has interference, the next hop moves to a clean frequency.
- Used in: Original 802.11 (1997) standard.
- Advantage: Highly resistant to narrowband jamming and interception.
- Disadvantage: Lower throughput compared to DSSS; not used in modern Wi-Fi.

Key Difference: DSSS spreads across all frequencies simultaneously using coding; FHSS hops between frequencies over time.

3. How do modulation schemes work in the PHY layer? Compare different modulation schemes and their performance across various Wi-Fi standards.

Answer:

Modulation is the process of encoding digital data onto a radio carrier wave by varying its amplitude, phase, or frequency. Higher-order modulation schemes encode more bits per symbol, increasing data rates but requiring a stronger signal (higher SNR).

Modulation	Bits/Symbol	Required SNR	Used In	Notes
BPSK	1	Low	802.11a/b/g/n/ac/ax (low rates)	Most robust, lowest speed
QPSK	2	Low-Medium	802.11a/b/g/n/ac/ax	2x BPSK efficiency
16-QAM	4	Medium	802.11a/g/n/ac/ax	Good balance of speed and reliability
64-QAM	6	Medium-High	802.11n/ac/ax	Used in 802.11n and later
256-QAM	8	High	802.11ac/ax	Introduced in Wi-Fi 5
1024-QAM	10	Very High	802.11ax (Wi-Fi 6)	Requires excellent signal quality
4096-QAM	12	Extremely High	802.11be (Wi-Fi 7)	Maximum throughput, needs near-perfect SNR

How QAM Works: QAM (Quadrature Amplitude Modulation) varies both the amplitude and phase of the signal. A 256-QAM constellation has 256 distinct points, each representing 8 bits. The AP and client negotiate the highest MCS (Modulation and Coding Scheme) that the signal quality allows.

Key Insight: Higher modulation = more bits per symbol = higher speed, but only possible at close range or with excellent signal quality.

4. What is the significance of OFDM in WLAN? How does it improve performance?

Answer:

OFDM stands for Orthogonal Frequency Division Multiplexing. It is the core modulation technique used in 802.11a/g/n/ac/ax and is the reason modern Wi-Fi can achieve high data rates reliably.

How OFDM Works:

OFDM divides a wide channel into many narrow, orthogonal subcarriers (e.g., 52 subcarriers in a 20 MHz channel). Each subcarrier is modulated independently with BPSK, QPSK, or QAM. The subcarriers are orthogonal to each other, meaning they don't interfere even though they overlap in frequency.

Significance and Performance Improvements:

- **Multipath Resistance:** In indoor environments, signals reflect off walls and arrive at the receiver at different times (multipath). OFDM's narrow subcarriers have a long symbol duration, making them far less sensitive to multipath interference than single-carrier systems like DSSS.
- **High Spectral Efficiency:** Multiple subcarriers carry data simultaneously, making full use of the available bandwidth.
- **Scalability:** OFDM supports variable modulation per subcarrier, allowing adaptation to channel conditions.
- **Guard Interval:** A cyclic prefix (guard interval) is added to each symbol to absorb multipath echoes without causing inter-symbol interference (ISI).
- **Foundation for OFDMA:** OFDM was extended to OFDMA in 802.11ax, allowing multiple users to share subcarriers simultaneously.

Impact: OFDM enabled 802.11a/g to reach 54 Mbps vs 11 Mbps for 802.11b (DSSS), and it remains the foundation of all modern Wi-Fi standards.

5. How are frequency bands divided for Wi-Fi? Explain different bands and their channels.

Answer:

Wi-Fi operates in three main frequency bands: 2.4 GHz, 5 GHz, and 6 GHz (Wi-Fi 6E and Wi-Fi 7). Each band has different characteristics and is divided into channels.

2.4 GHz Band:

- **Range:** 2.400 – 2.4835 GHz
- **Total Channels:** 14 (varies by country; US uses 1–11)
- **Non-overlapping Channels:** Only 3 (Channels 1, 6, 11) with 20 MHz width
- **Characteristics:** Longer range, better wall penetration, but more congested (used by microwaves, Bluetooth, baby monitors)
- **Standards:** 802.11b/g/n/ax

5 GHz Band:

- **Range:** 5.150 – 5.850 GHz
- **Total Channels:** Up to 25 non-overlapping 20 MHz channels (varies by region and DFS rules)
- **Channel Widths:** 20, 40, 80, 160 MHz
- **Characteristics:** Shorter range, less congestion, higher throughput
- **Standards:** 802.11a/n/ac/ax
- **DFS Channels:** Some channels (52–140) require Dynamic Frequency Selection to avoid radar interference

6 GHz Band (Wi-Fi 6E / Wi-Fi 7):

- **Range:** 5.925 – 7.125 GHz
- **Total Channels:** 59 non-overlapping 20 MHz channels (up to 7 × 160 MHz channels)
- **Characteristics:** Least congested, highest throughput, shortest range
- **Standards:** 802.11ax (Wi-Fi 6E), 802.11be (Wi-Fi 7)
- **Only newer devices support this band**

Key Principle: Higher frequency = shorter range but more available spectrum and less interference. Lower frequency = longer range but more crowded spectrum.

6. What is the role of Guard Intervals in WLAN transmission? How does a short Guard Interval improve efficiency?

Answer:

What is a Guard Interval (GI)?

A Guard Interval is a short period of silence inserted between OFDM symbols during transmission. Its purpose is to allow multipath reflections from the previous symbol to die out before the next symbol starts, preventing Inter-Symbol Interference (ISI).

Why It Is Needed:

In real-world environments, a transmitted signal bounces off walls, floors, and ceilings. These reflections arrive at the receiver slightly later than the direct signal. If the next symbol starts before the delayed reflection finishes, the receiver gets confused — this is ISI. The guard interval absorbs this delay spread.

Standard vs Short Guard Interval:

Guard Interval	Duration	Used In	Overhead	Best For
Long GI (Standard)	800 ns	All 802.11 standards	Higher overhead	Environments with high multipath (indoor, many obstacles)
Short GI	400 ns	802.11n, ac, ax	Lower overhead (~11% gain)	Clean environments with low multipath (open spaces, line-of-sight)

How Short GI Improves Efficiency:

- Reducing the GI from 800 ns to 400 ns means less time is wasted between symbols.
- This results in approximately 11% higher throughput with the same modulation and coding scheme.
- For example, 802.11n with 800 ns GI gives 65 Mbps per stream; with 400 ns short GI, it gives 72.2 Mbps per stream.
- Short GI should only be used when the environment has low multipath delay spread — otherwise ISI can increase error rates.

7. Describe the structure of an 802.11 PHY layer frame. What are its key components?

Answer:

An 802.11 PHY layer frame (called a PPDU – Physical Protocol Data Unit) consists of three main components: the PLCP Preamble, PLCP Header, and PSDU (MAC frame payload).

1. PLCP Preamble:

- Used for synchronization between transmitter and receiver.
- Allows the receiver to detect the start of a frame, synchronize its clock, and estimate the channel.
- Consists of Short Training Fields (STF) and Long Training Fields (LTF).
- STF: Used for signal detection, automatic gain control (AGC), and coarse frequency offset correction.
- LTF: Used for fine frequency offset correction and channel estimation.

2. PLCP Header (Signal Field):

- Contains information the receiver needs to decode the rest of the frame.
- Includes: Data Rate (MCS index), Frame Length (number of bytes in PSDU), and coding information.
- Always transmitted at the lowest mandatory rate (e.g., 6 Mbps BPSK in 802.11a) so all devices can read it.
- In newer standards (HT, VHT, HE), additional SIG fields carry MIMO, channel width, and spatial stream info.

3. PSDU – PHY Service Data Unit (Data Field):

- The actual MAC layer frame (MPDU) being transmitted.
- Encoded using the modulation and coding scheme indicated in the PLCP header.
- May contain padding bits to align to OFDM symbol boundaries.

Summary: Preamble → Synchronize | Header → Decode parameters | Data → Actual payload. This structure ensures reliable frame detection and decoding across all devices in the BSS.

8. What is the difference between OFDM and OFDMA?

Answer:

Feature	OFDM	OFDMA
Full Form	Orthogonal Frequency Division Multiplexing	Orthogonal Frequency Division Multiple Access
Used In	802.11a/g/n/ac	802.11ax (Wi-Fi 6) and later
Channel Usage	All subcarriers assigned to one user at a time	Subcarriers divided into Resource Units (RUs) shared among multiple users simultaneously
Efficiency	One user occupies entire channel per transmission	Multiple users transmit at the same time on different RUs
Latency	Higher — users must wait for their turn	Lower — concurrent transmissions reduce wait time
Best For	High-throughput single-user scenarios	Dense environments with many simultaneous users (stadiums, offices, airports)
Overhead	Less scheduling complexity	Requires more sophisticated scheduling at AP

Analogy:

OFDM is like a highway where only one car (user) can travel at a time — it gets the full road. OFDMA is like a highway with multiple lanes — many cars travel simultaneously, each in their own lane (Resource Unit). OFDMA dramatically improves performance in high-density deployments by reducing the time each device has to wait.

9. What is the difference between MIMO and MU-MIMO?

Answer:

Feature	MIMO	MU-MIMO
Full Form	Multiple Input Multiple Output	Multi-User Multiple Input Multiple Output

Feature	MIMO	MU-MIMO
Introduced In	802.11n (Wi-Fi 4)	802.11ac Wave 2 (Wi-Fi 5) – downlink; 802.11ax (Wi-Fi 6) – uplink + downlink
How It Works	Multiple antennas on AP and client used to send multiple spatial streams to ONE device	Multiple antennas on AP used to send separate streams to MULTIPLE devices simultaneously
Users Served	One user at a time (single-user)	Multiple users at the same time
Streams	Up to 4 spatial streams (802.11n), 8 (802.11ac)	Up to 4 users simultaneously (802.11ac), 8 users (802.11ax)
Benefit	Increases throughput for a single device	Increases overall network capacity in multi-device environments
Limitation	Other clients must wait for current transmission to finish	Requires client capability and AP to perform beamforming and scheduling

Key Insight:

MIMO improves speed for one device at a time. MU-MIMO serves multiple devices at once. In a home with 2-3 devices, MIMO is sufficient. In an office or school with dozens of devices, MU-MIMO is essential for maintaining performance.

10. What are PPDU, PLCP, and PMD in the PHY layer?

Answer:

PPDU – Physical Protocol Data Unit:

The PPDU is the complete frame format at the PHY layer. It is what actually gets transmitted over the air. It consists of the PLCP Preamble, PLCP Header, and the PSDU (the MAC data). Every Wi-Fi transmission on the medium is a PPDU.

PLCP – Physical Layer Convergence Protocol:

The PLCP is the upper sublayer of the PHY layer. It acts as an interface between the MAC layer and the physical medium. Its responsibilities are:

- Framing: Adds preamble and header to the MAC data (MPDU) to form the PPDU.
- Rate Negotiation: Signals the data rate, channel width, and MCS to the receiver in the PLCP header.
- Clear Channel Assessment (CCA): Monitors the medium and tells the MAC layer whether the channel is busy or free (used in CSMA/CA).
- Provides a standard interface so the same MAC can work with different physical mediums.

PMD – Physical Medium Dependent:

The PMD is the lower sublayer of the PHY layer. It handles the actual transmission and reception of bits over the radio medium. Its responsibilities are:

- Modulation and Demodulation: Converts digital bits into analog radio signals (and back).
- Encoding: Applies forward error correction (FEC) such as convolutional coding or LDPC.
- Frequency and Power: Controls the transmit frequency, power level, and antenna operations.
- The PMD sublayer is what differs between 802.11a (OFDM at 5 GHz), 802.11b (DSSS at 2.4 GHz), etc.

Summary: PPDU = the full transmitted frame. PLCP = prepares and frames the data, manages CCA. PMD = actually sends/receives the bits over the air using radio signals.

11. What are the types of PPDU? Explain the PPDU frame format across different Wi-Fi generations.

Answer:

Different 802.11 standards define different PPDU formats, each adding fields to support new capabilities like MIMO, wider channels, and multi-user transmission.

1. Non-HT PPDU (Legacy) – 802.11a/b/g:

- Format: Preamble (STF + LTF) → Signal Field (SIGNAL) → Data
- Supports up to 54 Mbps (802.11a/g) or 11 Mbps (802.11b)
- Single spatial stream only, no MIMO

2. HT PPDU (High Throughput) – 802.11n (Wi-Fi 4):

- Format: L-STF → L-LTF → L-SIG → HT-SIG → HT-STF → HT-LTF(s) → Data
- Adds HT-SIG field to signal MIMO parameters (number of spatial streams, channel width, MCS index)
- Supports up to 600 Mbps with 4 spatial streams and 40 MHz channels
- Two sub-types: HT-Mixed (backward compatible) and HT-Greenfield (no legacy support)

3. VHT PPDU (Very High Throughput) – 802.11ac (Wi-Fi 5):

- Format: L-STF → L-LTF → L-SIG → VHT-SIG-A → VHT-STF → VHT-LTF(s) → VHT-SIG-B → Data
- VHT-SIG-A: Carries MCS, number of streams, channel width (up to 160 MHz)
- VHT-SIG-B: Per-user information for MU-MIMO
- Supports up to 3.5 Gbps with 8 spatial streams

4. HE PPDU (High Efficiency) – 802.11ax (Wi-Fi 6):

- Format: L-STF → L-LTF → L-SIG → RL-SIG → HE-SIG-A → HE-SIG-B → HE-STF → HE-LTF(s) → Data
- HE-SIG-B: Carries per-user Resource Unit (RU) allocation for OFDMA
- Four sub-types: HE SU (single user), HE MU (multi-user OFDMA), HE TB (trigger-based uplink), HE ER SU (extended range)
- Supports up to 9.6 Gbps

5. EHT PPDU – 802.11be (Wi-Fi 7):

- Adds Multi-Link Operation (MLO) support
- Supports 320 MHz channels and 4096-QAM
- Up to 46 Gbps theoretical throughput

12. How is the data rate calculated?

Answer:

The data rate in Wi-Fi is not a fixed value — it is dynamically calculated based on several physical layer parameters. The formula for data rate is:

Data Rate = (Subcarriers × Bits/Symbol × Coding Rate × Spatial Streams) / Symbol Duration

Parameters Explained:

- Number of Data Subcarriers: Depends on channel width (e.g., 20 MHz OFDM → 48 data subcarriers; 40 MHz → 108 subcarriers).
- Bits per Symbol (Modulation): BPSK=1, QPSK=2, 16-QAM=4, 64-QAM=6, 256-QAM=8, 1024-QAM=10.
- Coding Rate (FEC Rate): Forward Error Correction adds redundancy. Common rates: 1/2, 2/3, 3/4, 5/6. A rate of 3/4 means 75% of bits are data, 25% are error correction.

- Spatial Streams (MIMO): Each additional spatial stream multiplies the data rate (e.g., 2 streams = 2× rate).
- Symbol Duration: Typically 4 μs in 802.11a/g/n/ac (3.2 μs OFDM + 0.8 μs GI); 12.8 μs in 802.11ax (with 0.8/1.6/3.2 μs GI options).

Example Calculation – 802.11n, 20 MHz, 64-QAM, 3/4 coding, 1 stream, short GI:

- Data subcarriers: 52
- Bits per subcarrier: 6 (64-QAM)
- Coding rate: 3/4
- Spatial streams: 1
- Symbol duration: 3.6 μs (with short GI)
- Data rate = $(52 \times 6 \times 3/4 \times 1) / 3.6 \mu s = 234 / 3.6 = 65 \text{ Mbps}$

MCS Index:

In practice, the combination of modulation + coding rate is represented as an MCS (Modulation and Coding Scheme) index (e.g., MCS 0 = BPSK 1/2, MCS 7 = 64-QAM 5/6 in 802.11n). The AP and client negotiate the highest MCS that the current SNR can support, and this is what determines the actual link speed shown in the Wi-Fi connection properties.